Math Refresher for Masters

Homework 5

1 (a) $\lim_{x\to\infty} \frac{x^2-2x+5}{7x+1} = \lim_{x\to\infty} \frac{x-2+\frac{5}{x}}{7+\frac{1}{x}} = \boxed{\infty}$ ($x\to\infty$, $\frac{5}{x}\to 0$, $\frac{1}{x}\to 0$)

(b) $\lim_{x\to\infty}\left(\sqrt{x^2+2x+3}-\sqrt{x^2-1}\right) = \lim_{x\to\infty}\left(\frac{\left(\sqrt{x^2+2x+3}-\sqrt{x^2-1}\right)\left(\sqrt{x^2+2x+3}+\sqrt{x^2-1}\right)}{\sqrt{x^2+2x+3}+\sqrt{x^2-1}}\right) = \lim_{x\to\infty}\left(\frac{x^2+2x+3-x^2+1}{\sqrt{x^2+2x+3}+\sqrt{x^2-1}}\right) =$

$= \lim_{x\to\infty}\left(\frac{2x+4}{\sqrt{x^2+2x+3}+\sqrt{x^2-1}}\right) = \lim_{x\to\infty}\left(\frac{2+\frac{4}{x}}{\sqrt{1+\frac{2}{x}+\frac{3}{x^2}}+\sqrt{1-\frac{1}{x^2}}}\right) = \frac{2}{1+1} = \boxed{1}$ ($\frac{4}{x}\to 2$, $\frac{2}{x}+\frac{3}{x^2}\to 1$, $\frac{1}{x^2}\to 1$)

(c) $\lim_{x\to 0}\left(\sqrt{x^2+2x+3}-\sqrt{x^2+1}\right) = \sqrt{0+0+3}-\sqrt{0+1} = \boxed{\sqrt{3}-1}$

(d) $\lim_{x\to\infty} \frac{x^2+x+1}{2x^2+2x-1} = \lim_{x\to\infty} \frac{1+\frac{1}{x}+\frac{1}{x^2}}{2+\frac{2}{x}-\frac{1}{x^2}} = \boxed{\frac{1}{2}}$

(e) $\lim_{x\to 0} \frac{x^2+1}{2x^2-2} = \frac{0+1}{0-2} = \boxed{-\frac{1}{2}}$

(f) $\lim_{x\to 0} \frac{x^2+x}{2x^2-x} = \lim_{x\to 0} \frac{x+1}{2x-1} = \boxed{-1}$

(g) $\lim_{x\to 1} \frac{x^2-3x+2}{x^2-5x+4} = \lim_{x\to 1} \frac{(x-1)(x-2)}{(x-1)(x-4)} = \lim_{x\to 1} \frac{x-2}{x-4} = \frac{1-2}{1-4} = \boxed{\frac{1}{3}}$

2 (a) $f(x) = x^2-x+1$

$f'(x) = \boxed{2x-1}$

(b) $f(x) = x^2\cdot e^x\cdot \sin x$

$f'(x) = (x^2e^x)'\cdot\sin x + x^2\cdot e^x\cdot\cos x = \boxed{xe^x(2\sin x + x\sin x + x\cos x)}$

(c) $f(x) = \ln(x^2+1)$

$f'(x) = \frac{2x}{|x^2+1|} = \boxed{\frac{2x}{x^2+1}}$

(d) $f(x) = \sin(x^2+1)$

$f'(x) = \boxed{2x\cos(x^2+1)}$

(e) $f(x) = e^{x^2+1}$

$f'(x) = \boxed{2x\cdot e^{x^2+1}}$

(f) $f(x) = \sqrt{x^2+1}$

$f'(x) = \frac{2x}{2\sqrt{x^2+1}} = \boxed{\frac{x}{\sqrt{x^2+1}}}$

(g) $f(x) = (\sin x)^{\ln x}$ $\quad \ln f(x) = \ln(\sin x)^{\ln x} = \ln x\cdot\ln\sin x$

$(\ln f(x))' = \frac{f'(x)}{f(x)}$

$f'(x) = f(x)\cdot(\ln f(x))' = \boxed{(\sin x)^{\ln x}\cdot\left(\frac{1}{|x|}\cdot\ln\sin x + \frac{\cos x}{|\sin x|}\cdot\ln x\right)}$

3 Prove by definition: $f(x)=x^2$ is uniformly continuous on $[0;1]$.

□ $\forall \varepsilon>0 \ \exists \delta>0 \ \forall x,y\in[0,1]: |x-y|<\delta \Rightarrow |f(x)-f(y)|<\varepsilon$

$|f(x)-f(y)| = |x^2-y^2| = |x-y||x+y|$

$x,y\in[0,1] \Rightarrow 0\le x+y\le 2 \Rightarrow |x^2-y^2| \le 2|x-y| < \varepsilon$

Take $\delta=\frac{\varepsilon}{2}$: $|f(x)-f(y)| \le 2|x-y| < 2\delta=\varepsilon$ ■

4 $\sin(3.1)$ - ?

$f(x)\approx f(a)+f'(a)(x-a)$

Let $a=\pi$ (because $3.1\approx\pi$)

$\sin(3.1)\approx \sin(\pi)+\cos(\pi)(3.1-\pi) = 0-1\cdot(3.1-\pi)=\pi-3.1\approx$ **0.04159**

5 $f(x)=2x^3-9x^2+12x+1$

$f'(x)=6x^2-18x+12=6(x^2-3x+2)=6(x-1)(x-2)$



$\Rightarrow$

$x=1$, $f(1)=2-9+12+1=6$ - point of local maximum

$x=2$, $f(2)=2\cdot8-9\cdot4+12\cdot2+1=5$ - point of local minimum

increasing: $(-\infty;1)\cup(2;+\infty)$

decreasing: $(1;2)$

6 $f(x)=x^3-2x^2+x+1$ $\quad x_0=2$ $\quad \lambda=0.1$

$x_{k+1}=x_k-\lambda\cdot f'(x_k)$ $\quad f'(x)=3x^2-4x+1$

1) $f'(2)=3\cdot4-4\cdot2+1=12-8+1=5$ $\quad x_1=2-0.1\cdot5=$ **1.5**

2) $f'(1.5)=3\cdot2.25-4\cdot1.5+1=1.75$ $\quad x_2=1.5-0.1\cdot1.75=$ **1.325**